

Experiment 15

Understand the data preprocessing in Data Science.

Program:

```
import numpy as np
import pandas as pd

df=pd.read_csv("/content/pre-process_datasample.csv")
df
df.info()

df.Country.fillna(df.Country.mode()[0],inplace=True)
df.Age.fillna(df.Age.median(),inplace=True)
df.Salary.fillna(round(df.Salary.mean()),inplace=True)
df

pd.get_dummies(df.Country)

updated_dataset=pd.concat(
    [pd.get_dummies(df.Country),df.iloc[:,[1,2,3]]],axis=1
)

updated_dataset

df.info()

updated_dataset.Purchased.replace(['No','Yes'],[0,1],inplace=True)
updated_dataset
```

Output

```
Output:
Country  Age  Salary  Purchased
0 France   44.0  72000.0    No
1 Spain    27.0  48000.0    Yes
2 Germany  30.0  54000.0    No
3 Spain    38.0  61000.0    No
4 Germany  40.0  63778.0    Yes
5 France   35.0  58000.0    Yes
6 Spain    38.0  52000.0    No
7 France   48.0  79000.0    Yes
8 France   50.0  83000.0    No
9 France   37.0  67000.0    Yes

After encoding Country and Purchased:
France Germany Spain Age Salary Purchased
0 True False False 44.0 72000.0 0
1 False False True 27.0 48000.0 1
2 False True False 30.0 54000.0 0
3 False False True 38.0 61000.0 0
4 False True False 40.0 63778.0 1
5 True False False 35.0 58000.0 1
6 False False True 38.0 52000.0 0
7 True False False 48.0 79000.0 1
8 True False False 50.0 83000.0 0
9 True False False 37.0 67000.0 1
```